

Paper I v1.3 Internal Audit Final Report

Internal author-side audit — not external

21 August 2026 — Overall verdict: PASS

Protocol: INTERNAL_AUDIT_STANDARD_v1.3.

Lean execution: recorded full build and external clean-room result reviewed; Lean not rerun.

1 Frozen target

Item	Value
English manuscript SHA-256	f3094b670c93ff622c3f573cdab61bdd0f5d84007f04b1888b364e0183565bea
Spanish manuscript SHA-256	57ba967fc2de805b7bbb4cf5f937727bde43c2ada80fa794bcbb5a727db05b8b
English PDF SHA-256	7d04c47692b613c8d6e2cc4471f0205128b4ae06ca11d581a08d020eb2236db0
Spanish PDF SHA-256	3ad16ae86e8fcad358bf964a6ae98ae053b0bde2fdf3140e008cedab78b90c2a
Formal archive	PAPER_I_lean_v1.2_freeze.zip
Formal archive SHA-256	0181506408644fc1f8872d711de5a98a500f4052aa295bcd6f8c82776694fd3a
Lean / Mathlib	Lean 4.28.0 / Mathlib 8f9d9cff6bd728b17a24e163c9402775d9e6a365
Recorded build / axioms	exit 0; 8,034 jobs; propext, Classical.choice, Quot.sound

2 Gate results

Gate	Verdict	Principal evidence
G0	PASS	95 static, integrity and residual controls; six LF-sealed artifacts.
G1	PASS after correction	Eight external findings resolved; one residual citation reference found internally, corrected and rerun.
G2	PASS	5 identities; 561 boundary; 480 orbit; 99,671 assembly; 1,179 split-LP; 9 sharpness checks.
G3	PASS	Headline, assembly and residual-duality surfaces match the unchanged archive.
G4	PASS recorded build	8,034-job record and 15-surface axiom report reviewed; no Lean rerun.
G5	PASS	English/Spanish aligned; known-block controls and general exact/near-duplicate scan pass.

G6	PASS	19/20 pages; clean logs, embedded fonts and 39 pages visually inspected.
G7	PASS internal	Primary CEO scan added; scoped novelty wording; specialist gate open.
G8	PASS	Self-contained v1.3 target, correction matrix and evidence perimeter.

3 Correction disposition

The v1.3 manuscript resolves all eight findings in the external v1.2 report: Spanish duplication, stale integrity baseline, exposition of (4.7), the domain of Appendix A.2, the scope of “sharp,” CEO primary-source access, the Schrijver pinpoint, and the unused repository reference. During the internal rerun, P1-IA-V13-001 found one remaining textual occurrence of the unverified Schrijver pinpoint. It was corrected in both languages and formats; the PDFs, hashes and all downstream gates were regenerated and rerun.

These corrections do not alter a theorem, hypothesis, numerical constant, proof step or Lean source. The formal archive is byte-identical to the archive already rebuilt independently by the external auditor.

4 Overall verdict

PASS. All nine blocking internal gates pass. There are no unresolved internal blockers or major findings. Residual external adversarial validation, independent specialist prior-art review, human peer review, tagging and publication approval remain open external gates.